

from price to performance. Processors are constantly evolving to meet the rapidly evolving applications, increasing memory, speed requirements, and reduced power consumption needs.

A smart card's components need to be very secure to overcome challenges of being hacked. This is being done by roping together several technologies including high-level ones like biometric identification. A secure processor core is usually tamper-resistant and prevents retrieval or modification of on-chip information even during the most sophisticated hacking activity. "Today, integrated circuits (ICs) that are used in smart cards have more memory capacities and are capable of performing multiple applications," says P.V. Bhas.

Radio frequency identity (RFID) tags and trusted platform module (TPM) technologies are also in the market today. "The fast growth of this sector indicates that trends of contactless smart cards will grow, supporting interfaces such as universal serial bus (USB), wireless USB, etc. Quick adaptation of technology other than smart card, such as NFC, reader NFC, and TPM, secured USB token for secured Internet transaction, and machine to machine (M2M) communication for various industrial applications, are the latest technological developments in this area," says Deepak Agarwal. STMicroelectronics offers a full range of secure ICs and advanced platforms for smartcard and digital security applications.

"Today, semiconductor suppliers are designing the chip using standard secured ARM cores instead of proprietary cores, which will result in standard operating system (OS) offerings from third parties. Other than this, there have been improvements in performance and power efficiency as we shift to 90nm and beyond," adds Deepak Agarwal.

E-purse, which is a new segment, is being explored in the industry, and if the technology succeeds it has huge potential. E-purses are issued by banks

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to their account holders. Money is loaded into the e-purse by a transfer from the card-holder's bank account at a load device. The card-holder's account is immediately debited and the value loaded into the e-purse is transferred to a float account at the bank or

the systems operator. This transaction is online and PIN-protected, just like a cash withdrawal from an ATM. The e-purse card-holder goes to the point of sale location where the e-purse is accepted. During e-payment, the value is transferred from the e-purse to the terminal, and value remaining in e-purse is adjusted. This is an offline transaction and the technology is yet to be widely adopted.

According to Mohsin Syed, vice president, sales and marketing, Infronics Systems, some other prevalent technologies that relate to and complement smart security technologies include RFID, secure networks, trusted software platforms, biometrics, memory cards and nanotechnologies. Infronics Systems specialises in industry specific high-quality software and hardware solutions in the core technology areas of RFID, smart cards, biometrics, enterprise mobility and embedded design and hardware production.

ON Semiconductor offers a comprehensive portfolio of smart card interface integrated circuits (ICs) covering all application ranges including point-of-sale (POS) payment terminals, set-top boxes and SIM interfaces. A smart card interface is basically a circuit that adapts or level-shifts the logic levels between a controller and a smart card IC. Additionally, it provides to the smart card the 1.8V, 3V or 5V power supply. As an interface it protects the application circuit or the upstream microcontroller.

"The interface can just be a basic three- or five-channel level shifter providing the power supply to the card or can include fault detection, card detection and protection functions through the addition of a core circuit block, which is the ISO7816 sequencer," explains Philippe Andre, applications engineer for signal and interface, ON Semiconductor. These applications are all based on the original ISO7816 smart card standard but with their specific standard derivatives and requirements. "The EMV standard is specifically dedicated for payment applications.